

Exploratory Data Analysis - projectA

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Setup and Libraries

Data Description

In this section, we load the Project A dataset and preview its structure. The dataset contains continuous metrics across two experimental groups: Control and Treatment A.

```
if (file.exists("project_a_data.csv")) {  
  df <- read.csv("project_a_data.csv")  
} else {  
  set.seed(42)  
  df <- data.frame(  
    ID = 1:100,  
    Group = factor(rep(c("Control", "Treatment A"), each = 50)),  
    Value_A = c(rnorm(50, mean = 50, sd = 10), rnorm(50, mean = 65, sd = 12)),  
    Value_B = c(rnorm(50, mean = 100, sd = 15), rnorm(50, mean = 130, sd = 20))  
  )  
}  
  
head(df)
```

```
##   ID   Group Value_A Value_B  
## 1   1 Control  63.70958 118.01448  
## 2   2 Control  44.35302 115.67127  
## 3   3 Control  53.63128  84.95187  
## 4   4 Control  56.32863 127.72723  
## 5   5 Control  54.04268  89.99840  
## 6   6 Control  48.93875 101.58271
```

Summary Statistics

In this section, we compute key summary statistics for each treatment group.

```
summary_tbl <- df %>%  
  group_by(Group) %>%  
  summarise(  
    Count = n(),  
    Mean_Value_A = round(mean(Value_A), 2),  
    SD_Value_A = round(sd(Value_A), 2),  
    Mean_Value_B = round(mean(Value_B), 2),
```

```
SD_Value_B = round(sd(Value_B), 2)
)

knitr::kable(summary_tbl, caption = "Summary Statistics by Group")
```

Table 1: Summary Statistics by Group

Group	Count	Mean_Value_A	SD_Value_A	Mean_Value_B	SD_Value_B
Control	50	49.64	11.51	97.73	13.91
Treatment A	50	66.21	11.10	129.53	17.70

Data Visualizations

In this section, we visualize the distribution and relationship of our key metrics.

```
ggplot(df, aes(x = Group, y = Value_A, fill = Group)) +
  geom_boxplot(alpha = 0.7, outlier.shape = NA) +
  geom_jitter(width = 0.2, alpha = 0.5, color = "black") +
  theme_minimal() +
  labs(
    title = "Distribution of Value A Across Treatment Groups",
    x = "Group",
    y = "Value A Metric"
  ) +
  theme(legend.position = "none")
```

```
ggplot(df, aes(x = Value_A, y = Value_B, color = Group)) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", se = FALSE, linetype = "dashed") +
  theme_minimal() +
  labs(
    title = "Scatter Plot of Value A vs. Value B",
    x = "Value A",
    y = "Value B",
    color = "Treatment Group"
  )
```

Observations

Based on the exploratory data analysis: * **Group Differences:** Treatment A demonstrates higher values for both Value A and Value B compared to the Control group. * **Positive Correlation:** There is a positive linear relationship between Value A and Value B across both groups.

Software Environment

To ensure reproducibility, the software environment parameters and package versions used to generate this document are recorded below:

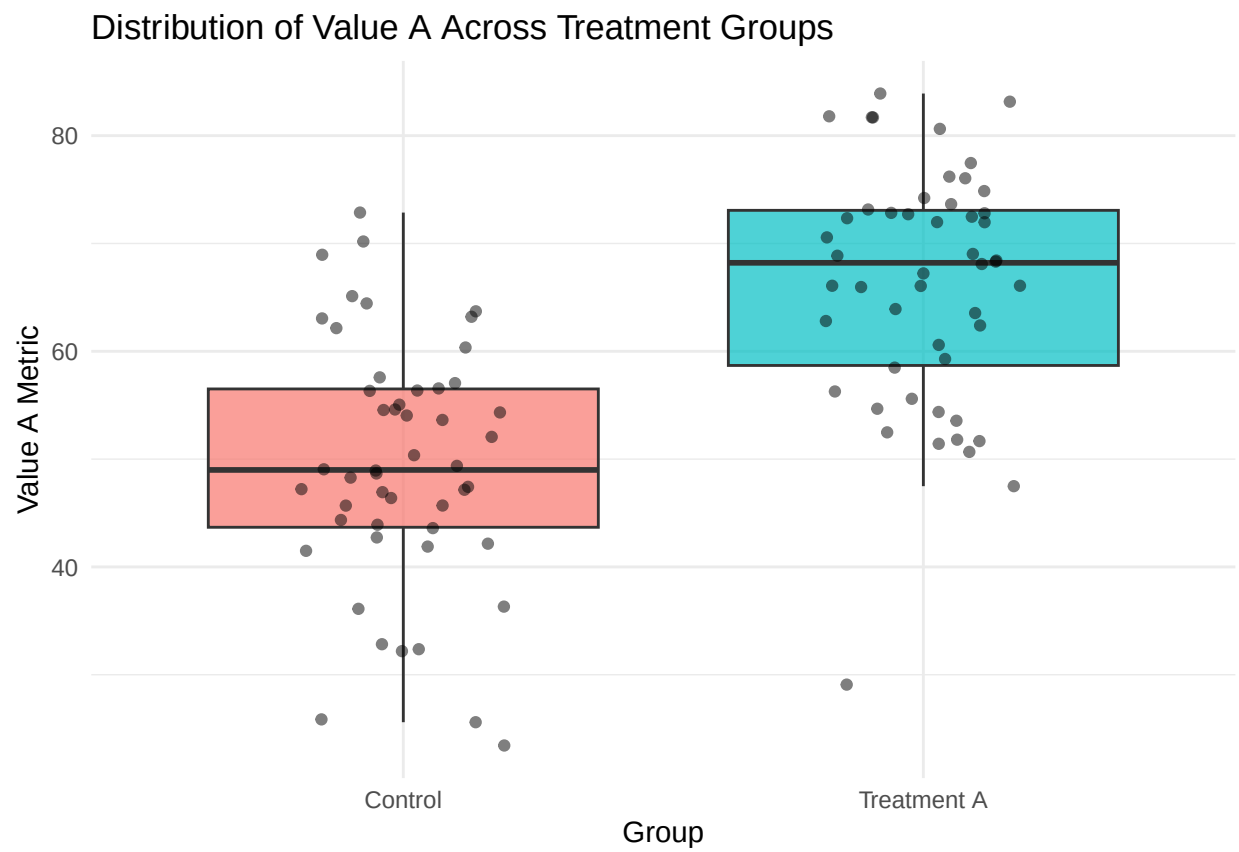


Figure 1: Figure 1: Distribution of Value A by Group



Figure 2: Figure 2: Relationship between Value A and Value B

```
sessionInfo()
```

```
## R version 4.6.1 (2026-06-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 22621)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United Kingdom.utf8
## [2] LC_CTYPE=English_United Kingdom.utf8
## [3] LC_MONETARY=English_United Kingdom.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United Kingdom.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] lubridate_1.9.5 forcats_1.0.1  stringr_1.6.0  dplyr_1.2.1
## [5] purrr_1.2.2     readr_2.2.0    tidyr_1.3.2    tibble_3.3.1
## [9] ggplot2_4.0.3   tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] Matrix_1.7-5      gtable_0.3.6      compiler_4.6.1    tidyselect_1.2.1
## [5] splines_4.6.1     scales_1.4.0      yaml_2.3.12       fastmap_1.2.0
## [9] lattice_0.22-9    R6_2.6.1          labeling_0.4.3    generics_0.1.4
## [13] knitr_1.51        pillar_1.11.1     RColorBrewer_1.1-3 tzdb_0.5.0
## [17] rlang_1.3.0       stringi_1.8.7     xfun_0.60         S7_0.2.2
## [21] ote1_0.2.0        timechange_0.4.0  cli_3.6.6         mgcv_1.9-4
## [25] withr_3.0.3       magrittr_2.0.5    digest_0.6.39     grid_4.6.1
## [29] rstudioapi_0.19.0 hms_1.1.4         nlme_3.1-169      lifecycle_1.0.5
## [33] vctrs_0.7.3       evaluate_1.0.5    glue_1.8.1        farver_2.1.2
## [37] rmarkdown_2.31    tools_4.6.1       pkgconfig_2.0.3   htmltools_0.5.9
```